

Experiment - 8.1.1. Multiplication of Two Square Matrices

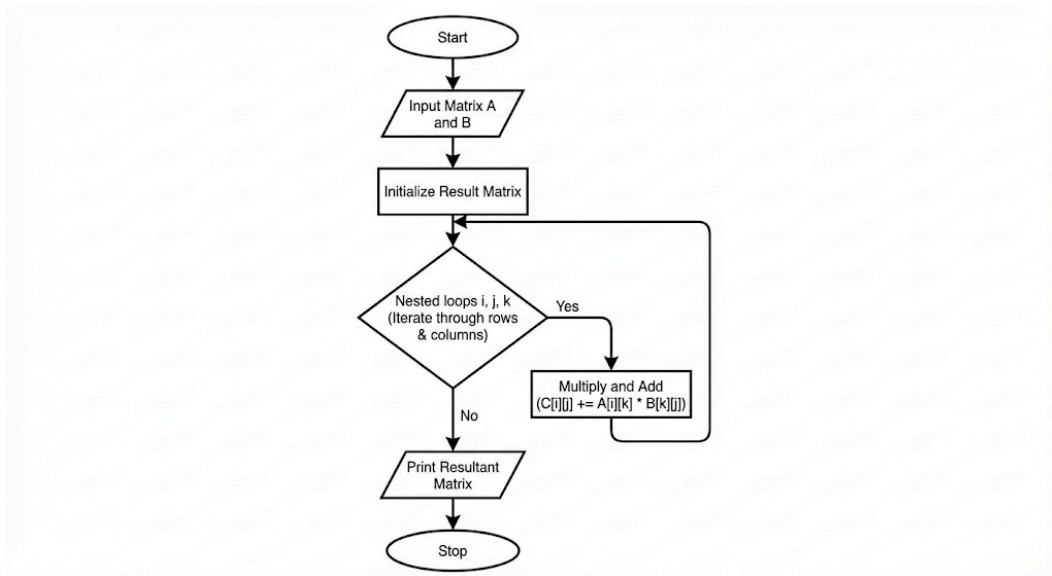
1. Aim

To design and implement a Python program that performs the multiplication of two square matrices of dimension $n \times n$. The program takes the dimension and the elements of both matrices as input from the user and computes their product.

2. Pseudocode

1. **START**
2. **PRINT** "dimension: " and **READ** an integer from the user, storing it in the variable n.
3. **PRINT** "first matrix:" and **INITIALIZE** an empty list A.
4. **FOR** each row i from 0 to n-1:
 - **READ** space-separated integers from the user, convert them into a list, and append to A.
5. **PRINT** "second matrix:" and **INITIALIZE** an empty list B.
6. **FOR** each row i from 0 to n-1:
 - **READ** space-separated integers from the user, convert them into a list, and append to B.
7. **INITIALIZE** a result matrix of size $n \times n$ filled with zeros.
8. **FOR** i in range n (iterating through rows of A):
 - **FOR** j in range n (iterating through columns of B):
 - **FOR** k in range n (iterating through rows of B):
 - **ADD** the product of A[i][k] and B[k][j] to result[i][j].
9. **PRINT** "Resultant Matrix:".
10. **FOR** each row r in the result matrix:
 - **PRINT** the elements of r separated by spaces.
11. **END**

3. Flowchart



4. Python Program

```
n = int(input("dimension: "))

print("first matrix:")
A = []
for i in range(n):
    A.append(list(map(int, input().split()))))

print("second matrix:")
B = []
for i in range(n):
    B.append(list(map(int, input().split()))))

result = [[0]*n for _ in range(n)]

# Matrix multiplication logic
for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j] += A[i][k] * B[k][j]

print("Resultant Matrix:")
for r in result:
    print(*r)
```

5. Experiment Screenshot

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8.1.1. Multiplication of Two Square Matrices

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n , representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases

```
1 n = int(input("dimension: "))
2
3 print("first matrix:")
4 A = []
5 for i in range(n):
6     A.append(list(map(int, input().split())))
7
8 print("second matrix:")
9 B = []
10 for i in range(n):
11     B.append(list(map(int, input().split())))
12
13 result = [[0]*n for _ in range(n)]
14
15 for i in range(n):
16     for j in range(n):
```

Average time: 0.009 s (9.50 ms) Maximum time: 0.012 s (12.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 0 ms

Expected output

dimension: 2

first matrix:

1 2

3 4

second matrix:

5 6

Terminal Test Cases

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